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COI Name (Institution):	Toburen, Larry (East Carolina University)		
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Task Description:

The interpretation of data from mechanistic studies on radiation biology requires detailed descriptions of the local patterns of energy deposition in the appropriate cellular or sub-cellular sites to explore the effects of radiation quality on biological response. The objective of this project is to provide Monte Carlo (MC) simulations on the local patterns of energy deposition by HZE particles in the appropriate cellular environment including effects of micro-, and macro-geometrical considerations. The calculations focus on interpretation of microscopic “event” or “dose” distributions appropriate to recently initiated studies of HZE radiation induced leukemia in animal and cellular models [NSCOR Program, Colorado State University, R. Ulrich, PI] and those encountered in the production of clustered damage in DNA [e.g., Brookhaven National Laboratory, B Sutherland, PI]. The progenitor cells for leukemia are haematopoietic stem cells that are located in clusters within marrow cavities having dimensions on the order of a few hundred microns. These cavities are surrounded by trabecular bone matrix, which forms a wall of several hundred microns thickness around the marrow. The dimensions and geometry of the bone marrow are too irregular for direct dosimetry measurements and dose-averaging methods are inappropriate for chronic low dose-rate (single particle) exposures. Detailed Monte Carlo charged particle track structure simulation will provide quantitative information on spatial patterns of energy deposition in cellular targets of dimensions as small as a few nanometers on a particle-by-particle basis; these distributions are the starting point to characterize the detailed nature of radiation-induced damage produced by high-LET radiation. The modeling approach will combine the capabilities of full event-by-event Monte Carlo track simulation codes with codes employing condensed history capabilities; the latter is more appropriate for the description of high-energy secondary electrons. The event-by-event ion code PROTRAC (which is part of the biophysical simulation tool PARTRAC) developed at GSF Neuherberg for protons and alpha particles will be extended for heavy charged particle (HZE) interactions using appropriate relativistic theory. Generated secondary electrons that traverse specific regions of interest, e.g., bone marrow cells or their nuclei, will be followed also event-by-event using ETRAC (the electron transport code of PARTRAC) whereas electron transport in adjoining tissues and the surrounding structural material will be followed by the more computer efficient condensed history general purpose MC transport code PENELOPE developed at the University of Barcelona. Appropriate interaction cross sections for a wide range of target materials will be added to the ETRAC and PROTRAC codes, specifically to explore the effects of bone-tissue interfaces on energy deposition in bone marrow cells. Comparison of the spatial pattern of energy deposited in volumes of interest obtained from models of the HZE particles with that obtained from low-LET radiation will provide a firm basis to interpret radiation quality effects on biological response and the mathematical structure developed is amenable to scaling to volumes appropriate to human dimensions for considerations of risk assessment.
The objective of this project is to provide Monte Carlo (MC) simulations on the local patterns of energy deposition by HZE particles in the appropriate cellular environment including effects of micro-, and macro-geometrical considerations. The focus will be on Leukemia, i.e., on the bone/bone-marrow interface from realistic mouse or human dimensions down to single cells.

Little is known on the interaction of HZE particles with the human body. HZE particles are part of the Galactic Cosmic Rays (GCR) which exist in deep space and cannot be shielded completely inside spacecrafts. The latency time of leukemia, 2-3 years, coincides more or less with the mission time for the planned expedition to Mars.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Work is currently in progress in several directions. We are working on the development and improvement of the PARTRAC module for the simulation of the HZE particles. This includes an update of the liquid water cross sections and the development of interaction cross sections for calcium, a major constituent of bone. We have developed and tested a realistic but generic bone/bone-marrow target model using the geometry package of PENELOPE. This generic model is based on experimental information of human trabecular bones from different bone sites. Details of the model are published in Gersh et al. Health Phys. 93 (2007) 47-59. We are currently working on extending the model to rodents. We have finished coupling the geometry model (PENELOPE) with the proton, alpha-particle and HZE track structure modules of PARTRAC. We have currently started simulations of energy depositions by different radiation qualities and energies within the human and rodent bone models. We are also working on models to characterize the influence of tissue inhomogeneity on marow dose in trabecular spongiosa.
HZE Track Structure Simulations:

We have developed an interaction model for bare (i.e., fully ionized) and sufficiently fast heavy ions by relating HZE interaction cross sections to ionization and excitation cross sections of protons by scaling laws. HZE inelastic interaction cross sections (total and energy differential) are obtained from the proton cross sections by energy scaling and multiplication by the ion charge squared. Secondary electron emission angles are modeled as a superposition of soft collisions (dipole interaction) and hard or binary collisions; they are calculated for each ion separately.

While liquid water serves as substitute for soft tissue and bone marrow, we will use metallic calcium, a major component of bone, as substitute for bone in PARTRAC. PENELOPE uses a sum of atomic cross sections due to the atomic constituents (ICRP recommendations) of cortical bone and bone marrow.

We are currently updating inelastic proton interaction cross sections with liquid water and have started the derivation of proton interaction cross sections with calcium. We have adopted a model for the dielectric response function of calcium based on theoretical models and available experimental information and are currently checking sum rules and other properties of the optical and dielectric functions. The extinction coefficient was adjusted in the energy range around 25 eV to account for the electrons in the calcium M shell. Sum rules are correctly obtained for all quantities, Kramers-Kronig relations calculated and checked. Currently, we are working on the extension algorithm to add momentum dependence to the dielectric response function. This is the first step in calculating interaction cross sections within the plane-wave Born approximation.

Realistic Bone/Bone-Marrow Model:

We have developed a simple, non-bone-site-specific, non-patient-specific, and computationally efficient geometrical model of trabecular spongiosa using the geometry package of the general-purpose MC transport code PENELOPE. Internal dimensions of the model are obtained from path-length distributions as measured in different human bones by research groups at the University of Florida and the University of Leeds. We have benchmarked and validated our model by determining path-length distributions and by calculating absorbed dose by internal monoenergetic electron-emitters and by comparing it to more sophisticated tomography-based and chord-length-based transport models

published in the literature.

We have started work to extend our geometrical model to murines. Internal dimensions of murine bones published in the literature strongly suggest that simple scaling of our human model suffice. Therefore, in a first step, our murine model (which we refer to as scaled mouse model consists of the human model scaled by a factor of 0.1. However, a thorough analysis of micro-CT scans of murine trabecula should be performed to obtain realistic internal dimensions and distributions comparable to those used in the development of the human model.

HZE Particle Transport through the Bone/Bone-Marrow Model:

We have successfully coupled the event-by-event code PARTRAC – which supports only slab-geometries – with our trabecular spongiosa geometry model. We have developed a method to reduce our realistic geometry used in PENELOPE to a simple slab geometry which can be used in PARTRAC. As a first step we plan to transport HZE particles with PARTRAC while secondary electrons are transported with PENELOPE. Currently, we are testing the new hybrid code by transporting alpha particles from external as well as internal emitters.

Characterizing energy deposition: the Marrow Dose Heterogeneity Factor (MDHF):

The Marrow Dose Heterogeneity Factor (MDHF) is a quantity used to describe the extent by which the microgeometry and tissue inhomogeneity of the spongiosa region influences the patterns of energy deposition by ionizing radiation within the confines of the marrow cavities of the trabecular spongiosa. The MDHF is defined as the ratio of the actual energy deposited in a region to the homogeneous deposition estimate for a region. If for instance the marrow cavity were subdivided into 10 regions of equal mass, the homogeneous deposition estimate would be determined by dividing the emission energy of the electrons emitted from within the region by the number of equally-massed concentrically-layered divisions, in this case 10. If electrons were deposit their energy in a truly homogeneous fashion, such as the case at low emission energies (e.g., 5 – 10 keV), each equally-massed region will receive the same amount of energy deposition. However an exception occurs; for the case of energy depositions within the closest-to-the-interface layer, since only one side of the layer is a source-emitting marrow and the other is simply bone, there will be less energy deposited in this layer than in the next-to-closest-to-the-interface layer.

Bibliography Type:	Description: (Last Updated: 11/09/2018)
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